INDIVIDUAL THESIS

GPS-BASED CUSTOMER BEHAVIOUR TRACKING AND REAL-TIME STAFF ASSISTANCE

An Individual Component of the Smart Omni-Retail System

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Individual component: AI-Based Virtual Try-On and Size Recommendation System

Group ID: R26-IT-060

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DECLARATION AND SUPERVISOR CERTIFICATION

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ABSTRACT

Physical apparel stores need to identify when shoppers may require help without relying entirely on manual observation or camera-based monitoring. This component proposes a mobile, location-aware customer behaviour tracking and real-time staff assistance system within an AI-Based Smart Retail Recommendation System for Apparel. The store is represented as zones, and a mobile client sends location updates to a FastAPI backend. Point-in-polygon checks associate updates with zones, a jitter filter reduces small location fluctuations, and a Random Forest classifier distinguishes browsing from transiting behaviour. Dwell-time and movement-pattern rules are used to identify possible assistance opportunities and send alerts to store staff through a dashboard. The supplied research paper reports 91.2% classification accuracy, weighted precision of 0.908, weighted recall of 0.912, weighted F1-score of 0.910, and ROC-AUC of 0.895. These results require confirmation against the final model output and evaluation protocol. The draft documents the problem, related work, research method, system design, implementation, testing, results, limitations, and future work. GPS accuracy in indoor retail environments, battery use, consent, and the risk of interpreting movement as a customer's emotional state remain important limitations.

Keywords: retail analytics; GPS; geofencing; point-in-polygon; customer behaviour; Random Forest; real-time staff assistance; mobile application; privacy.

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My sincere thanks also go to my fellow research group members, J.A.R.K. Wijetunga, T.A Silva, W.M Methsara for their collaboration, support, and contributions to the overall Smart Omni-Retail System, particularly during the integration of its different components.

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Table of Contents

DECLARATION AND SUPERVISOR CERTIFICATION 2
ABSTRACT 3
ACKNOLEDGEMENTS 4
List of Abbreviations 5
CHAPTER 1 — INTRODUCTION 12
1.1 Overall Project Context 12
1.2 Individual Component Context 13
1.3 Problem Statement 14
1.4 Research Gap 14
1.5 Aim and Objectives 15
1.6 Research Questions 15
1.7 Scope, Limitations and Assumptions 16
1.8 Individual Contribution 16
1.9 Thesis Structure 17
CHAPTER 2 — LITERATURE REVIEW AND RELATED WORK 17
2.1 Review Method and Scope 17
2.2 Domain and Theoretical Foundations 17
2.2.1 Location-based services and GPS 17
2.2.2 Geofencing and point-in-polygon 18
2.2.3 Dwell time and movement patterns 18
2.2.4 Supervised classification and Random Forest 18
2.2.5 Real-time dashboards and alert workflows 18
2.2.6 Privacy, consent and responsible use 18
2.3 Related Research 18
2.4 Existing Systems, Products and Techniques 19
2.5 Gap Synthesis and Conceptual Framework 20
2.6 Chapter Summary 21
CHAPTER 3 — RESEARCH METHODOLOGY 21
3.1 Research Design 21

3.2 Data, Participants and Study Materials 21
3.3 Experimental and Study Procedure 21
3.4 Measures and Evaluation Metrics 22
3.5 Data Analysis 23
3.6 Ethics, Privacy, Security and Responsible AI 23
3.7 Reproducibility and Research Artefacts 23
3.8 Chapter Summary 23
CHAPTER 4 — PRODUCT AND SOLUTION DESIGN 24
4.1 Requirements 24
4.2 Component Architecture Within the Complete Solution 25
4.3 Design Alternatives and Decisions 25
4.4 Detailed Design 26
4.4.1 Location update processing 26
4.4.2 Zone assignment and transition state 26
4.4.3 Behaviour classification and assistance rules 26
4.4.4 Data model and interfaces 27
4.5 Standards and Professional Practice 27
4.6 Chapter Summary 27
CHAPTER 5 — IMPLEMENTATION, INTEGRATION AND TESTING 28
5.1 Development Environment and Technology Stack 28
5.2 Component Implementation 28
5.2.1 Mobile location acquisition 28
5.2.2 Backend endpoint and validation 28
5.2.3 Zone engine and filtering 28
5.2.4 Random Forest and rule engine 29
5.2.5 Dashboard and alert resolution 29
5.3 Integration with Other Components 30
5.4 Verification and Software Testing 31
5.5 Deployment and Operation 31
5.6 Chapter Summary 31
CHAPTER 6 — RESULTS AND VALIDATION 32
6.1 Research Results 32
6.2 Product and Technical Results 33

6.3 User or Stakeholder Validation 34
6.4 Integration Results 34
6.5 Summary of Findings 34
6.6 Chapter Summary 34
CHAPTER 7 — DISCUSSION 35
7.1 Interpretation of Findings 35
7.2 Comparison with Literature, Baselines and Existing Solutions 35
7.3 Research Contribution 36
7.4 Product Contribution 36
7.5 Limitations and Threats to Validity 37
7.6 Implications 37
7.7 Chapter Summary 37
CHAPTER 8 — COMMERCIALISATION AND SUSTAINABILITY 38
8.1 Value Proposition and Target Users 38
8.2 Market and Competitive Position 38
8.3 Business or Sustainability Model 38
8.4 Scalability, Intellectual Property and Compliance 39
8.5 Contribution to Sustainable Development Goals 40
8.6 Chapter Summary 40
CHAPTER 9 — CONCLUSION AND FUTURE DIRECTIONS 41
9.1 Conclusion 41
9.2 Achievement of Objectives 41
9.3 Individual Research and Product Contributions 41
9.4 Future Directions 41
9.5 Final Summary 42
LIST OF PLANNED FIGURES 43
REFERENCES 44
GLOSSARY 45
APPENDICES 46
Appendix A — Dataset Card Template 46
Appendix B — API Documentation Template 46
Appendix C — Test Evidence 46
Appendix D — User Feedback Instrument 46

Appendix E — Model Configuration and Results 46
Appendix F — Figure Creation Checklist 47

LIST OF FIGURES

Figure 1.1: Overall Smart Retail System Context 11
Figure 1.2: Individual Component Boundary and Ownership 12
Figure 2.1: Conceptual Framework and Research Gap 19
Figure 3.1: Research Methodology Workflow 21
Figure 4.1: Component Architecture 24
Figure 4.2: Zone Transition and Assistance Decision Flow 25
Figure 5.1: Mobile-to-Backend Event Sequence 28
Figure 6.1: Confusion Matrix for Browsing versus Transiting 31
Figure 6.2: Classification Metric Comparison 32
Figure 7.1: Comparison of Positioning/Analytics Approaches 35
Figure 8.1: Deployment and Commercial Operating Model 38
Figure 8.2: Privacy-Aware Data Lifecycle 39
Figure 9.1: Future Work Roadmap 41

LIST OF TABLES

Table 1.1: Components of the Wider Smart Retail System 12
Table 1.2: Research Objectives and Evidence Required 14
Table 1.3: Scope, Limitations, and Assumptions 15
Table 1.4: Individual and Shared Contribution Boundary 15
Table 2.1: Suggested Literature Search Protocol 16
Table 2.2: Related-Work Comparison 18
Table 2.3: Technique-Level Comparison 18
Table 3.1: Dataset Documentation Checklist 20
Table 3.2: Evaluation Metrics and Definitions 21
Table 3.3: Reproducibility Artefacts 22
Table 4.1: Functional Requirements 23
Table 4.2: Non-Functional Requirements 23
Table 4.3: Design Alternatives 24
Table 4.4: Proposed Core Data Entities (Confirm Against Implementation) 26
Table 5.1: Technology Stack 27
Table 5.2: Integration Matrix 29
Table 5.3: Test Plan 30
Table 6.1: Classification Results Reported in the Supplied Research Paper 31
Table 6.2: Operational Results Template 32
Table 6.3: Feedback Summary from Component Explanation 33
Table 7.1: Evidence-Based Comparison Framework 34
Table 7.2: Limitations and Validity Threats 36
Table 8.1: Stakeholders and Value Proposition 37
Table 8.2: Indicative Cost and Revenue Categories 38
Table 8.3: Sustainability and SDG Mapping 39
Table 9.1: Objective Achievement Matrix 40

LIST OF ABBREVIATIONS

Abbreviation	Meaning
API	Application Programming Interface
AUC	Area Under the Curve
F1	Harmonic mean of precision and recall
GPS	Global Positioning System
HTTP	Hypertext Transfer Protocol
ML	Machine Learning
PII	Personally Identifiable Information
ROC	Receiver Operating Characteristic
SDK	Software Development Kit
UI	User Interface
UX	User Experience
XAI	Explainable Artificial Intelligence

1. INTRODUCTION

1.1 Overall Project Context

The wider AI-Based Smart Retail Recommendation System for Apparel combines several data-driven functions intended to improve the retail experience and operational decision-making. The supplied project presentation identifies four components: GPS-based customer behaviour tracking and staff assistance; smart inventory and stock-flow optimisation; AI-based virtual try-on and size recommendation; and personalized marketing intelligence. This thesis focuses on the GPS component. Other components should be described only to explain the wider platform and verified integration points.

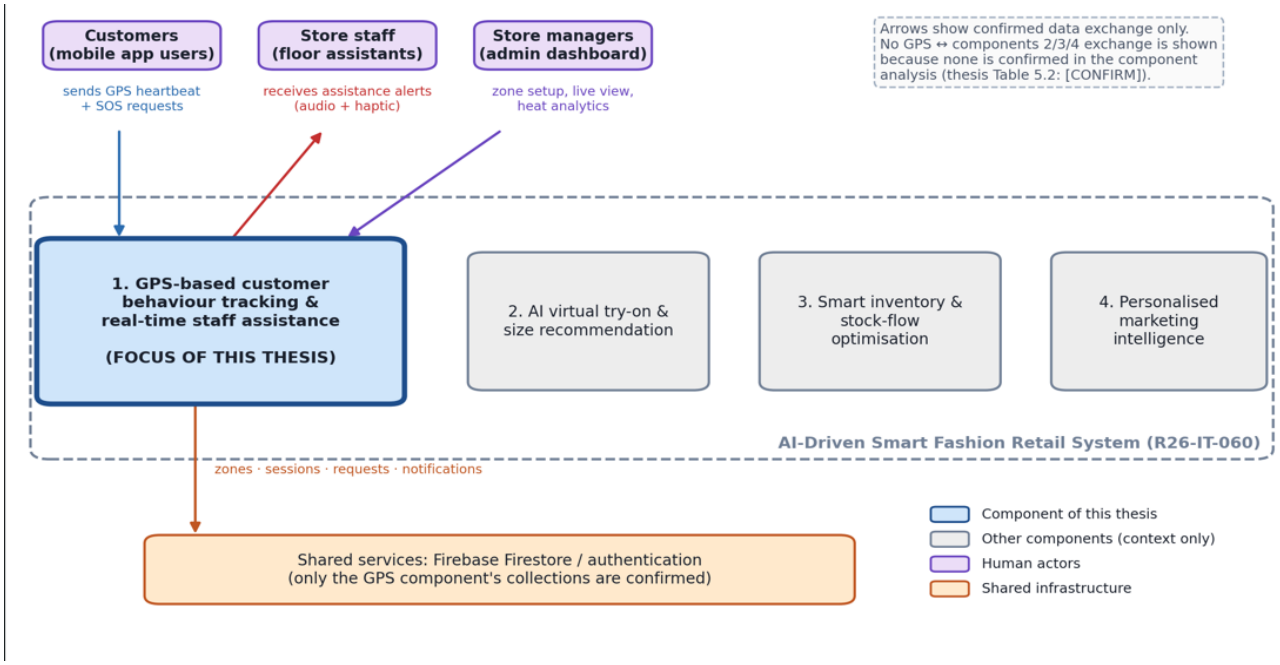


Figure 1.1: Overall smart retail system context

Table 1.1: Components of the wider smart retail system

No.	Component	Member	Student ID
1	GPS-based customer behaviour tracking and real-time staff assistance	L.A.D.A.S. Perera	IT22243812
2	AI-based virtual try-on and size recommendation	Silva T. A	IT22284952
3	AI-driven smart inventory and stock-flow optimisation	W.M Methsara	IT22244598
4	Personalized marketing intelligence	WIJETUNGA J A R K	IT22146588

1.2 Individual Component Context

The component is designed to track a customer's movement between predefined retail zones and identify movement patterns that may indicate a need for assistance. A mobile application obtains location updates, a backend maps coordinates to store zones, and behaviour logic classifies activity as browsing or transiting. Dwell and movement rules can then raise an assistance request for staff. A dashboard presents alerts and relevant zone/customer state. The system is intended to use common mobile hardware rather than dedicated cameras or beacons.

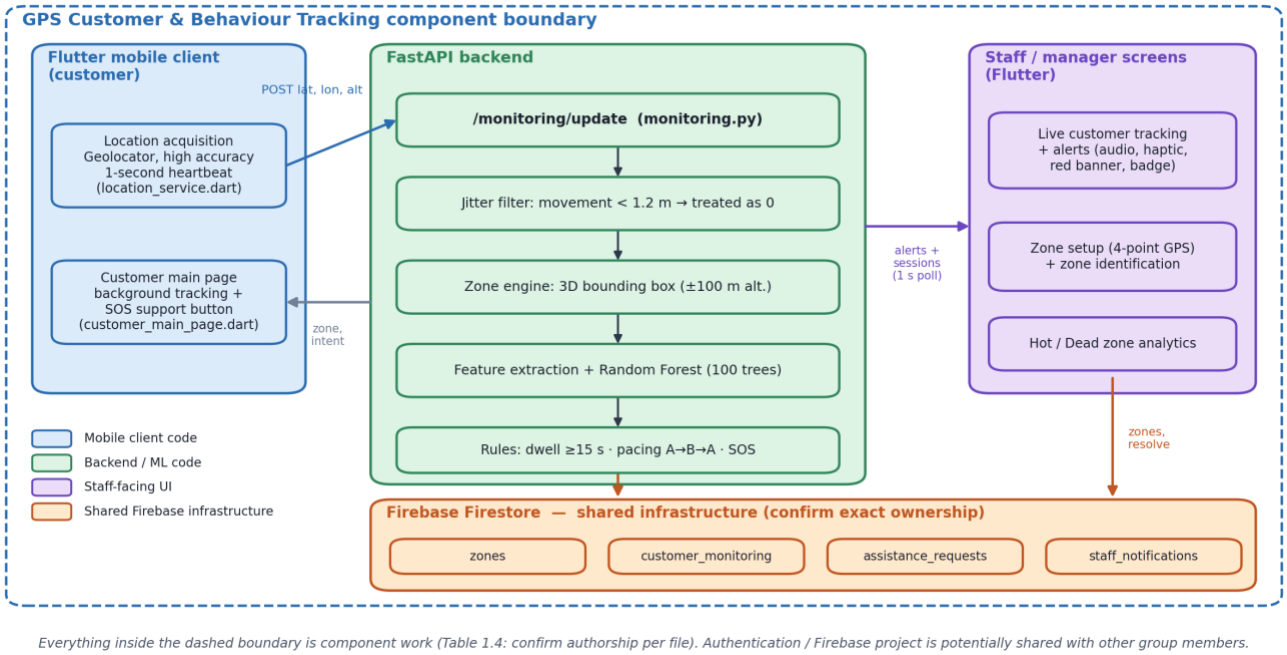


Figure 1.2: Individual component boundary and ownership

1.3 Problem Statement

Retail employees cannot continuously observe every shopper in a multi-zone store. Manual observation is time-consuming and may fail to identify customers who spend an unusually long time in an area or repeatedly move between nearby zones. Camera-based monitoring and dedicated indoor-positioning infrastructure can introduce cost, deployment, and privacy concerns. Without timely signals, staff may respond late to customers who need product guidance, potentially reducing service quality and shopping satisfaction.

The problem addressed by this component is how to use mobile location updates and lightweight behavioural classification to identify potential assistance opportunities and notify staff promptly, while acknowledging that GPS may be inaccurate indoors and that movement alone cannot establish a customer's intent or emotional state.

1.4 Research Gap

The supplied component explanation states that existing tracking systems commonly rely on cameras or sensors and that few systems combine customer behaviour tracking with real-time staff assistance in a low-cost and scalable approach. The project paper also notes indoor GPS degradation, battery use, background-app dependence, and the need for retraining with real-world movement data. The gap should therefore be framed as a practical integration and validation gap—not as a claim that no similar systems exist. Related literature already studies geofencing, dwell-time measurement, retail movement analysis, and location-aware services.

The proposed contribution is the integration of zone-based location processing, jitter tolerance, a lightweight Random Forest classifier, dwell/movement rules, and staff alerts in a mobile-to-backend workflow. The effectiveness of this combination must be evaluated with explicit classification and operational measures.

1.5 Aim and Objectives

Aim: To design, implement, and evaluate a GPS-based customer behaviour tracking component that identifies selected movement patterns and provides timely assistance alerts to retail staff.

Table 1.2: Research objectives and evidence required

Objective	Planned approach	Evidence needed
O1: Define store zones and detect current zones from location updates	Point-in-polygon processing and location filtering	Zone definitions, unit tests, boundary cases
O2: Classify movement as browsing or transiting	Random Forest model using documented trajectory features	Dataset description, split strategy, confusion matrix, precision/recall/F1
O3: Identify potential assistance opportunities	Dwell threshold and movement-pattern rules	Rule tests, false-positive/false-negative analysis
O4: Notify staff and maintain alert state	Dashboard and backend/Firestore workflow	End-to-end logs, alert latency, resolution-state tests
O5: Assess practical limitations	GPS accuracy, battery, latency, privacy review	Device tests, limitations, consent/security evidence

1.6 Research Questions

RQ1: How effectively can the selected movement features distinguish browsing from transiting behaviour in the evaluated dataset?

RQ2: How reliably does the zone and rule engine detect configured assistance-trigger patterns under location noise?

RQ3: What is the end-to-end delay between a qualifying event and its appearance on the staff dashboard?

RQ4: What privacy, usability, and deployment limitations arise when using GPS for in-store assistance?

1.7 Scope, Limitations and Assumptions

Table 1.3: Scope, limitations, and assumptions

Category	Description
In scope	Mobile location updates; zone detection; jitter filtering; browsing/transiting classification; dwell/movement rules; staff alerts; dashboard and resolution state
Out of scope unless confirmed	Camera-based emotion detection; dedicated beacon/UWB positioning; customer identity inference; production-wide live-store deployment
Assumptions	The customer grants required permissions; the app can obtain location updates; store zones are configured correctly; network connectivity is available for backend updates
Limitations	GPS drift indoors; device differences; background restrictions; battery consumption; movement is only a proxy and does not prove customer intent; limited or synthetic training data may not generalise
Evidence still needed	Actual dataset provenance/size, device test matrix, alert-latency logs, consent text, security review, live deployment status

1.8 Individual Contribution

This section must identify the author's own work separately from shared group work. Based on the component explanation, the scope may include mobile location collection, zone logic, behavioural rules, API integration, Random Forest classification, dashboard alerts, and evaluation. The final list must be reconciled against Git history, code ownership, presentation slides, and the actual repository.

Table 1.4: Individual and shared contribution boundary

Work area	Draft classification	Confirmation required
Flutter mobile location client	Component work	Confirm files/modules authored
FastAPI update endpoint and zone processing	Component/backend work	Confirm whether personally implemented or shared
Random Forest training/evaluation	Component ML work	Confirm model owner and experiment artefacts
Dashboard and alert workflow	Component work	Confirm UI modules and shared frontend services
Firebase/Firestore/authentication	Potential shared infrastructure	Identify exact shared services
System-wide integration	Group work	List actual integrations and responsible members

1.9 Thesis Structure

Chapter 1 defines the context, problem, gap, aim, questions, scope, and contribution. Chapter 2 reviews location tracking, geofencing, retail behaviour analytics, classification, real-time alerting, and privacy. Chapter 3 presents the research and evaluation methodology. Chapter 4 describes requirements and design. Chapter 5 documents implementation and testing. Chapter 6 reports verified results. Chapter 7 interprets findings and limitations. Chapter 8 discusses commercialisation and sustainability. Chapter 9 concludes and proposes future work.

2. LITERATURE REVIEW AND RELATED WORK

2.1 Review Method and Scope

The literature review should use peer-reviewed studies, standards, legislation, and credible system documentation. The current draft uses the supplied research paper and a small set of external contextual sources. Before submission, document the databases searched, keywords, publication-year range, inclusion/exclusion criteria, and how sources were screened. Do not describe this as a systematic review unless a reproducible systematic method was actually followed.

Table 2.1: Suggested literature search protocol

Element	Draft plan — confirm actual process
Search terms	GPS geofencing; retail customer tracking; dwell time; location-based services; browsing/transiting classification; staff assistance alerts; mobile location privacy
Sources	IEEE Xplore, ACM Digital Library, SpringerLink, ScienceDirect, Google Scholar, legislation and official documentation
Include	Peer-reviewed work on geofencing, movement analysis, indoor/outdoor positioning, retail analytics, privacy, and real-time systems
Exclude	Unverifiable claims, unrelated location applications, marketing pages used as scientific evidence
Screening record	Record date, query, number of results, duplicates, full-text decisions, and reason for exclusion

2.2 Domain and Theoretical Foundations

2.2.1 Location-based services and GPS

GPS provides geographic coordinates from a mobile device and is suitable for outdoor positioning, but accuracy may degrade inside buildings due to attenuation, reflections, and limited satellite visibility. The supplied research paper explicitly identifies indoor GPS accuracy as a limitation. Therefore, this system should treat coordinates as uncertain observations rather than exact ground truth. A location quality threshold, smoothing, stale-update handling, and explicit uncertainty states are important.

2.2.2 Geofencing and point-in-polygon

Geofencing defines a virtual boundary around a physical area. For a retail store, a floor plan can be divided into zones represented by polygons. A point-in-polygon test determines whether a coordinate lies within a zone. The design must specify coordinate reference assumptions, polygon boundaries, overlapping zones, edge cases, and behaviour when a point lies outside all configured zones.

2.2.3 Dwell time and movement patterns

Dwell time estimates how long a shopper remains in a zone or region. It can support operational decisions, but a long dwell does not necessarily mean confusion or a desire for help. The component explanation describes a dwell threshold greater than 15 seconds and a repeated A→B→A movement pattern within 60 seconds. These should be described as configurable heuristics and tested against alternative thresholds rather than treated as universally valid behavioural truths.

2.2.4 Supervised classification and Random Forest

Random Forest combines predictions from multiple decision trees and can model nonlinear relationships among movement features. It is a plausible lightweight classifier for structured features such as displacement, elapsed time, speed estimates, zone changes, and dwell duration. The thesis must list the exact features and labels used in training, describe how labels were obtained, and ensure that train/test splits do not leak segments from the same trajectory or person across sets.

2.2.5 Real-time dashboards and alert workflows

A real-time assistance workflow includes event detection, alert delivery, staff acknowledgement, and resolution. Operational quality is not captured by classification accuracy alone: alert delay, duplicate rate, false-alarm burden, resolution time, and availability also matter. Dashboard polling is described as one-second polling in the component summary; confirm whether this is the actual implementation and whether it creates acceptable load.

2.2.6 Privacy, consent and responsible use

Location data can reveal sensitive behavioural patterns. A responsible design should explain why location is collected, request permission, make tracking visible, limit collection to the relevant retail session, restrict staff access, avoid retaining raw trajectories longer than necessary, and provide deletion or retention controls where applicable. Sri Lanka's Personal Data Protection Act No. 9 of 2022 is relevant to the legal analysis; the current commencement and applicability of particular provisions must be verified from official sources before submission.

2.3 Related Research

Geofencing research discusses the design and evaluation of location boundaries and warns that detection sensitivity depends on location quality and time spent inside a region. Research on retail positioning and behaviour analytics has examined shopper paths, dwell time, and location-aware services using radio beacons, real-time location systems, and sensor fusion. These works establish that zone-based retail analysis is not new. The component's contribution should be described narrowly as its specific low-cost mobile-GPS pipeline and integrated assistance-alert workflow, subject to validation.

Table 2.2: Related-work comparison

Source/system	Main focus	Relevance	Limitation for this component
Supplied group research paper	GPS-based tracking, zone logic, Random Forest behaviour classification and staff alerts	Direct description of proposed component	Underlying dataset and experimental protocol need confirmation
Geofencing methodology research (Springer, 2023)	Methods and implementation issues in geofencing-based behavioural research	Supports discussion of boundary sensitivity and dwell time	Not specific to this exact apparel-store workflow
Retail positioning and sensor-fusion research (Pattern Recognition Letters, 2016)	Radio beacon and vision sensor fusion for retail profiling	Shows prior work on movement paths and customer profiling	Uses different sensing infrastructure; not equivalent to GPS-only design
RTLS shopper-path study (Journal of Retailing and Consumer Services)	Retail path segmentation and dwell/area analytics	Supports use of movement and dwell measures	Requires different positioning infrastructure and dataset
Indoor geofencing / navigation research (Scientific Reports, 2026)	Indoor localization, geofencing and ML-assisted navigation	Highlights indoor-positioning noise and privacy trade-offs	Different system, datasets, and objectives; not a direct performance benchmark

2.4 Existing Systems, Products and Techniques

Existing approaches include direct observation by staff, CCTV analytics, Bluetooth beacon systems, Wi-Fi fingerprinting, dedicated real-time location systems, and mobile geofencing. Manual observation is simple but inconsistent at scale. Camera analytics can provide richer visual context but brings infrastructure and privacy considerations. Beacons and Wi-Fi can support indoor positioning but require deployment and calibration. Mobile GPS can lower hardware cost but is less reliable indoors. The system therefore trades fine-grained indoor precision for accessibility and a simpler deployment model.

Table 2.3: Technique-level comparison

Technique	Potential strengths	Potential weaknesses	Implication
Manual observation	No new technical infrastructure	Does not scale; subjective; difficult to log consistently	Automated alerts may supplement staff judgement
CCTV analytics	Visual context and coverage	Cost, privacy, occlusion, camera placement	Avoid claiming equivalent visual understanding
BLE beacons / Wi-Fi positioning	Potentially better indoor location than GPS	Infrastructure, calibration, device variation	Possible future alternative or hybrid
GPS + geofencing	Common smartphone hardware; outdoor location support	Indoor drift and signal instability	Use uncertainty handling and cautious interpretation
GPS + ML + rules	Combines movement features and configurable triggers	Depends on labels, data quality and threshold tuning	Evaluate classification and alert workflow separately

2.5 Gap Synthesis and Conceptual Framework

The review indicates a practical opportunity to combine low-cost mobile location updates with zone-based event processing, movement classification, and a staff response workflow. The conceptual framework is: location observation → quality/jitter filtering → zone assignment → movement feature extraction → browsing/transiting classification → rule-based assistance trigger → staff notification → acknowledgement/resolution → privacy-aware event logging. The system must not equate a classifier label with a confirmed emotional state or a definitive request for help.

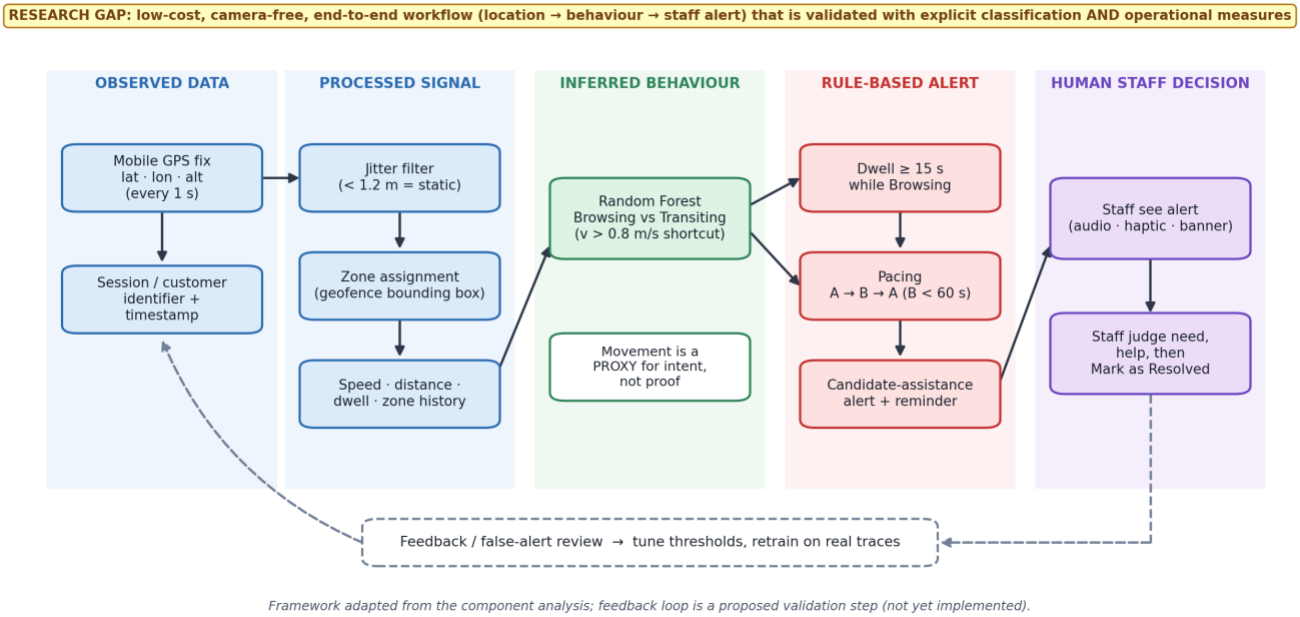


Figure 2.1: Conceptual framework and research gap

2.6 Chapter Summary

The literature supports the relevance of location-aware retail analytics but also shows that positioning accuracy, behavioural interpretation, and privacy are major concerns. The next chapter describes how the component can be evaluated transparently.

3. RESEARCH METHODOLOGY

3.1 Research Design

Use a design-and-evaluation approach: define requirements, implement a working prototype, evaluate the movement classifier on a documented dataset, test geofence/rule logic using controlled cases, and measure end-to-end alert behaviour. If no real customer study was conducted, state that evaluation is limited to offline data and controlled software tests rather than claiming real-world behavioural validity.

3.2 Data, Participants and Study Materials

The supplied paper reports a behaviour-classification dataset but the available excerpt does not provide sufficient provenance, sample size, collection protocol, label definition, or split details. These are essential to reproduce the results. The draft must not assume that the data were collected from real customers or from a live retail store.

Table 3.1: Dataset documentation checklist

Item	Required detail	Current status
Source/provenance	Collected, simulated, public, or synthetic; source and permissions	[CONFIRM]
Sample size	Number of trajectories, sessions, updates, users/devices	[CONFIRM]
Features	Exact columns, units, derived features and missingness	[CONFIRM]
Labels	Definition and annotation method for Browsing and Transiting	[CONFIRM]
Sampling	Update frequency, timestamp handling, session segmentation	[CONFIRM]
Split	Train/validation/test ratios and grouping strategy	[CONFIRM]
Privacy	Consent, anonymisation, retention, access restrictions	[CONFIRM]

3.3 Experimental and Study Procedure

A reproducible procedure should include: (1) define and validate store-zone polygons; (2) ingest and validate location updates; (3) filter jitter and stale/invalid observations; (4) map coordinates to zones; (5) construct movement features; (6) train and tune the Random Forest classifier; (7) evaluate on a held-out test set; (8) test dwell and repeated-zone rules; (9) send qualifying events through the API and dashboard; and (10) measure latency, failures, duplicate alerts, and resolution-state behaviour.

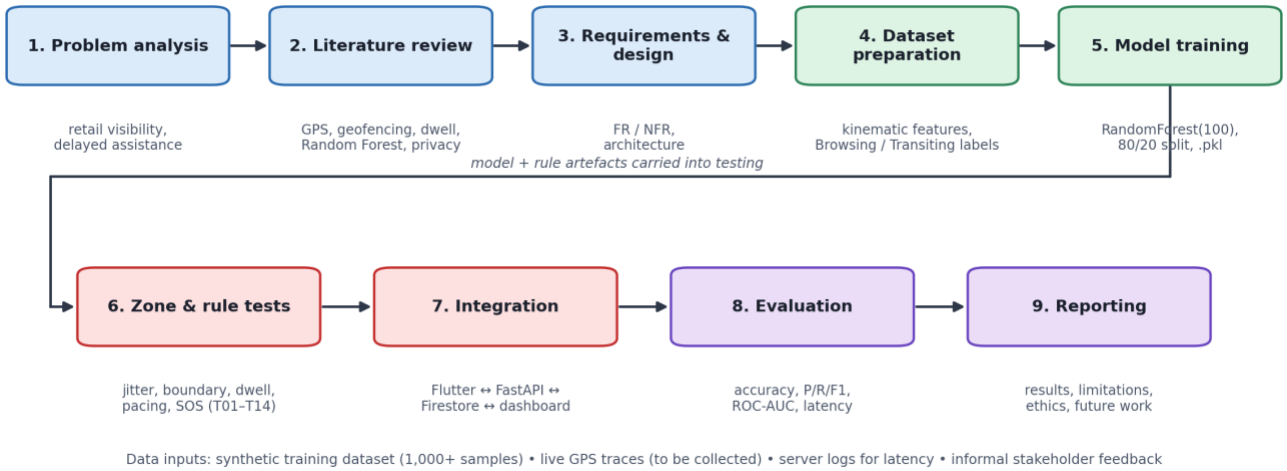


Figure 3.1: Research methodology workflow

3.4 Measures and Evaluation Metrics

For classification, report accuracy, per-class precision and recall, weighted F1, confusion matrix, and ROC-AUC only where probability scores and class definitions support it. If class imbalance exists, include macro-F1 and per-class support. For operational behaviour, measure event-detection precision/recall against manually labelled scenarios, alert latency, duplicate rate, API error rate, and resolution workflow success. Define each metric and the measurement unit.

Table 3.2: Evaluation metrics and definitions

Metric	Definition / use	Evidence required
Accuracy	Correct predictions divided by all predictions	Confusion matrix and test-set size
Precision	Of predicted instances of a class, proportion correct	Per-class counts
Recall	Of actual instances of a class, proportion detected	Per-class counts
Weighted F1	Class F1 values weighted by support	Exact calculation/output
ROC-AUC	Ranking ability across thresholds for binary/multiclass setup as configured	Probability outputs and evaluation method
Alert latency	Time from qualifying event to staff-visible alert	Timestamped logs and percentile summary
False-alert rate	False triggers divided by evaluated non-trigger opportunities	Annotated test scenarios
Resolution success	Share of alerts correctly acknowledged/resolved in workflow tests	Test cases and state logs

3.5 Data Analysis

Use a held-out test set and report raw counts alongside percentages. Compare the final model to a simple baseline where possible, such as a rule-only classifier or majority-class predictor. If thresholds such as 15 seconds and 60 seconds were selected empirically, document the tuning process and test alternative values. Do not tune thresholds on the final test set. Provide uncertainty estimates or repeated-run variability when sample size allows.

3.6 Ethics, Privacy, Security and Responsible AI

The system handles location data and can create inferences about customer behaviour. The evaluation should document consent and transparency, purpose limitation, least-privilege access, secure transport, authentication, retention limits, and deletion processes. Avoid identifying customers by name unless operationally necessary and approved. Staff alerts should be phrased as prompts to offer assistance, not as statements that a customer is confused. Review compliance with applicable Sri Lankan data-protection requirements and institutional ethics rules.

3.7 Reproducibility and Research Artefacts

Table 3.3: Reproducibility artefacts

Artefact	Expected contents	Status
Dataset card	Source, schema, collection, labels, exclusions, privacy	[MISSING/ADD]
Model configuration	Features, Random Forest parameters, random seed, library versions	[MISSING/ADD]
Evaluation outputs	Confusion matrix, classification report, ROC data, test counts	[MISSING/ADD]
Zone definitions	Coordinate reference, polygon vertices, boundary policy	[MISSING/ADD]
API contract	Request/response schema, authentication, error codes	[MISSING/ADD]
Test evidence	Automated test output, screenshots, logs, device details	[MISSING/ADD]

3.8 Chapter Summary

This methodology separates predictive classification performance from zone/rule correctness and end-to-end alert performance. Dataset provenance and reproducible evaluation are prerequisites for making strong claims.

4. PRODUCT AND SOLUTION DESIGN

4.1 Requirements

Table 4.1: Functional requirements

ID	Requirement	Priority	Acceptance evidence
FR-01	Obtain location updates after permission is granted	High	Permission and update tests
FR-02	Assign valid coordinates to configured zones	High	Inside/outside/boundary tests
FR-03	Filter small location fluctuations using documented jitter logic	High	Synthetic jitter tests
FR-04	Classify movement as Browsing or Transiting	High	Held-out model evaluation
FR-05	Apply dwell and repeated-zone rules to identify candidate assistance events	High	Rule unit tests
FR-06	Send event updates to backend endpoint	High	API integration tests
FR-07	Display current state and alerts on staff dashboard	High	UI/end-to-end evidence
FR-08	Allow staff to acknowledge and resolve alerts	Medium	State-transition tests
FR-09	Log minimal event data for audit and evaluation	Medium	Data-retention and log review

Table 4.2: Non-functional requirements

ID	Requirement	How to evaluate
NFR-01	Timely alerting	Measure event-to-dashboard latency
NFR-02	Reliability under missing/stale GPS	Fault injection and stale-update tests
NFR-03	Privacy and access control	Permission, authentication and retention review
NFR-04	Usability for staff	Task-based usability study if performed
NFR-05	Maintainability	Module boundaries, configuration and test coverage
NFR-06	Resource efficiency	Battery/network observation over a defined test session
NFR-07	Explainability	Expose the triggering zone/rule and not unsupported intent claims

4.2 Component Architecture Within the Complete Solution

The intended architecture contains a Flutter mobile client using Geolocator, a FastAPI endpoint identified in the component notes as /monitoring/update, a zone-processing and behaviour-analysis layer, a staff dashboard, and Firebase/Firestore state where implemented. The exact hosting, authentication, data schema, and dashboard technology need confirmation. Keep external services and shared infrastructure visually distinct from individual component code.

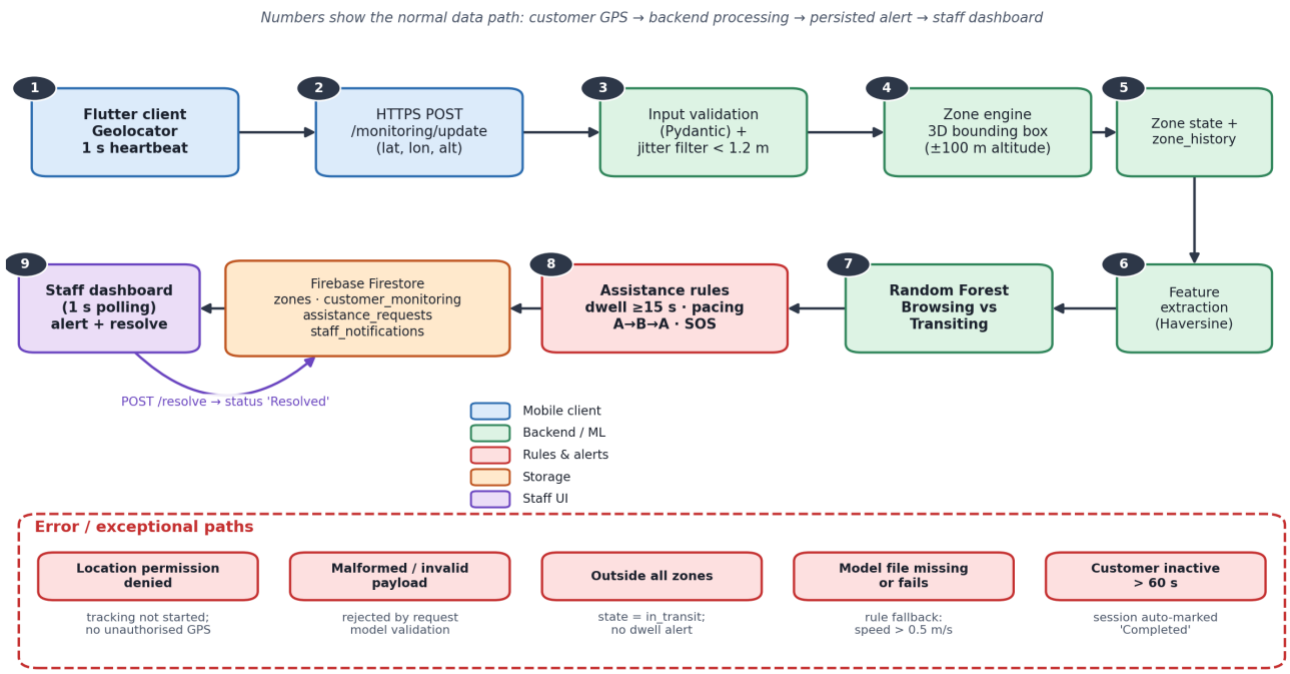


Figure 4.1: Component architecture

4.3 Design Alternatives and Decisions

Table 4.3: Design alternatives

Decision	Alternative A	Alternative B	Draft rationale / caveat
Location source	GPS on ordinary phones	BLE/Wi-Fi/UWB infrastructure	GPS lowers infrastructure requirements but indoor accuracy is weaker
Behaviour logic	Rules only	Random Forest plus rules	ML may capture patterns; rules remain transparent and configurable
Zone assignment	Circular radius	Polygon zones	Polygons can better represent aisles/areas but need accurate configuration
Alert delivery	Polling dashboard	Push/websocket events	Polling is simpler; confirm actual design and scalability
Tracking analysis	Camera analytics	Mobile location signals	Mobile signals avoid camera hardware but cannot infer visual context

4.4 Detailed Design

4.4.1 Location update processing

Validate the payload, timestamp, coordinate range, device/session identifier, and permission state. Reject malformed or stale observations. Apply the configured jitter filter before assigning zones. The 1.2 m jitter threshold appears in the component summary and must be verified against source code and measurement units. If GPS uncertainty exceeds a defined threshold, mark the location as uncertain rather than forcing a confident zone.

4.4.2 Zone assignment and transition state

Represent each zone using a polygon and stable zone identifier. Point-in-polygon processing determines zone membership. Define how the system handles overlapping polygons, shared boundaries, no-zone results, GPS jumps, and repeated events. A transition record should include timestamp, previous zone, current zone, and confidence/quality flags where available.

4.4.3 Behaviour classification and assistance rules

The Random Forest model is reported to classify Browsing versus Transiting. Document each feature and its calculation. Rules described in the component summary include dwell time greater than 15 seconds and an A→B→A sequence within 60 seconds. The alert rule should be explainable and configurable, and it should produce a candidate-assistance signal rather than a diagnosis of confusion.

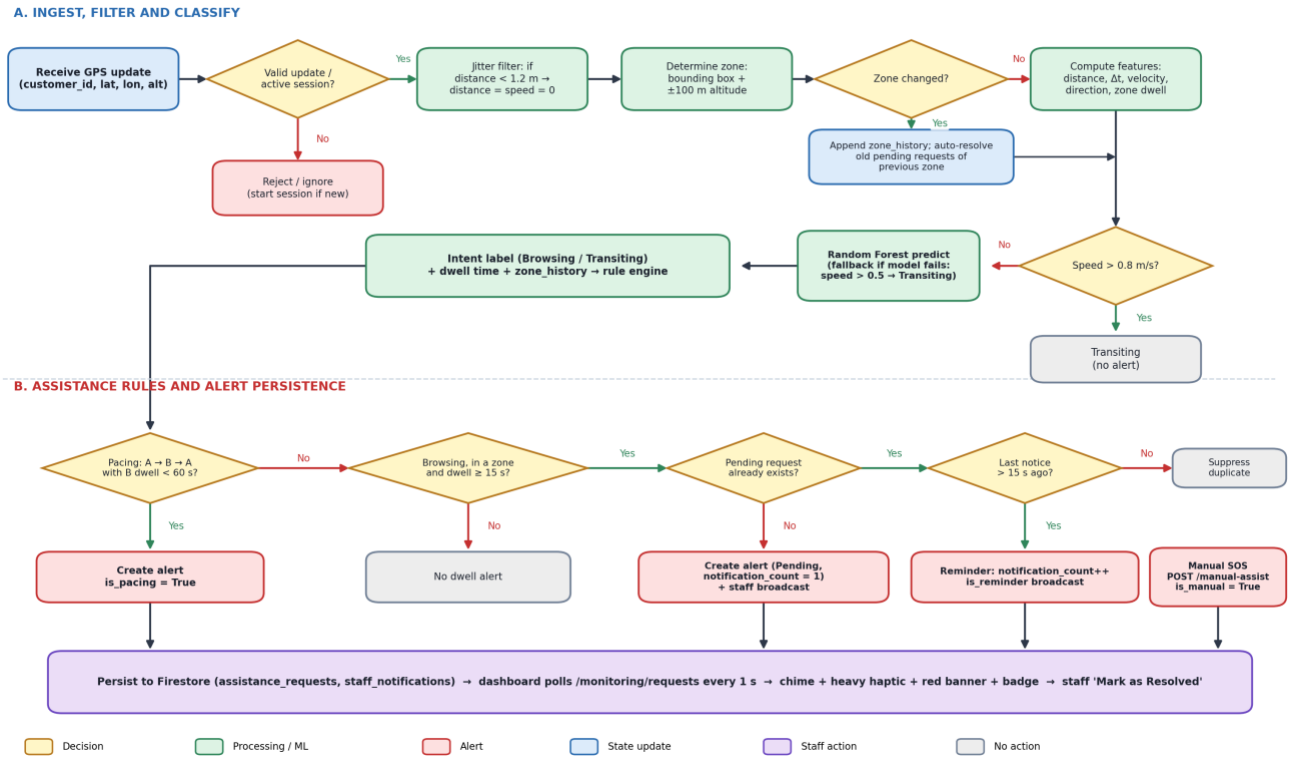


Figure 4.2: Zone transition and assistance decision flow

4.4.4 Data model and interfaces

Table 4.4: Proposed core data entities (confirm against implementation)

Entity	Example fields	Purpose
StoreZone	zone_id, store_id, polygon, label, enabled	Defines monitored areas
LocationEvent	session_id, timestamp, latitude, longitude, accuracy	Represents a location observation; minimise retention
ZoneTransition	session_id, from_zone, to_zone, timestamp	Supports movement features
BehaviourPrediction	session_id, label, model_version, score	Stores classifier output if implemented
AssistanceAlert	alert_id, trigger_rule, zone_id, created_at, status	Tracks staff-facing assistance events
AlertAction	alert_id, staff_role/id, action, timestamp	Records acknowledgement/resolution

4.5 Standards and Professional Practice

Use secure transport, input validation, access control, error handling, logging without unnecessary location exposure, versioned APIs, tests for boundary conditions, and documented model limitations. The exact technical standards and institutional requirements should be added after reviewing the actual deployment and supervisor guidance.

4.6 Chapter Summary

The design balances low-cost mobile sensing with explicit uncertainty handling, interpretable zone rules, and a human-in-the-loop staff response. The architecture must be updated to match the actual code rather than the proposed design alone.

5. IMPLEMENTATION, INTEGRATION AND TESTING

5.1 Development Environment and Technology Stack

Table 5.1: Technology stack

Layer	Technology stated in supplied material	Confirmation needed
Mobile client	Flutter	Flutter version, packages and device OS
Location API	Geolocator / Android location permissions	Package version and background behaviour
Backend	FastAPI	Python version, hosting and API schema
Endpoint	/monitoring/update	Request/response examples and auth
Zone processing	Point-in-polygon and jitter filtering	Library/algorithm and threshold units
ML	Random Forest classifier	Library, parameters, feature list, model serialization
Data/state	Firebase / Firestore	Collections, security rules, retention
Dashboard	Staff dashboard; exact framework unspecified	Framework, polling mechanism, screenshot
Notifications	Audio and haptic cues reported in component summary	Confirm exact channels and permissions
Version control/deployment	[CONFIRM]	Repository, CI/CD, host, environment configuration

5.2 Component Implementation

5.2.1 Mobile location acquisition

The mobile client requests the appropriate Android permissions, including ACCESS_FINE_LOCATION and INTERNET as described in the component notes. The implementation should handle denied permissions, disabled location services, reduced accuracy, app lifecycle changes, network loss, and battery constraints. Include screenshots of permission prompts and the location/tracking state only if they are from the real application.

5.2.2 Backend endpoint and validation

The endpoint /monitoring/update is identified in the component summary as the update route. Document its HTTP method, authentication, payload schema, response schema, validation errors, and rate-limiting behaviour from the actual API. Do not publish credentials, access tokens, real customer coordinates, or sensitive logs in the thesis.

5.2.3 Zone engine and filtering

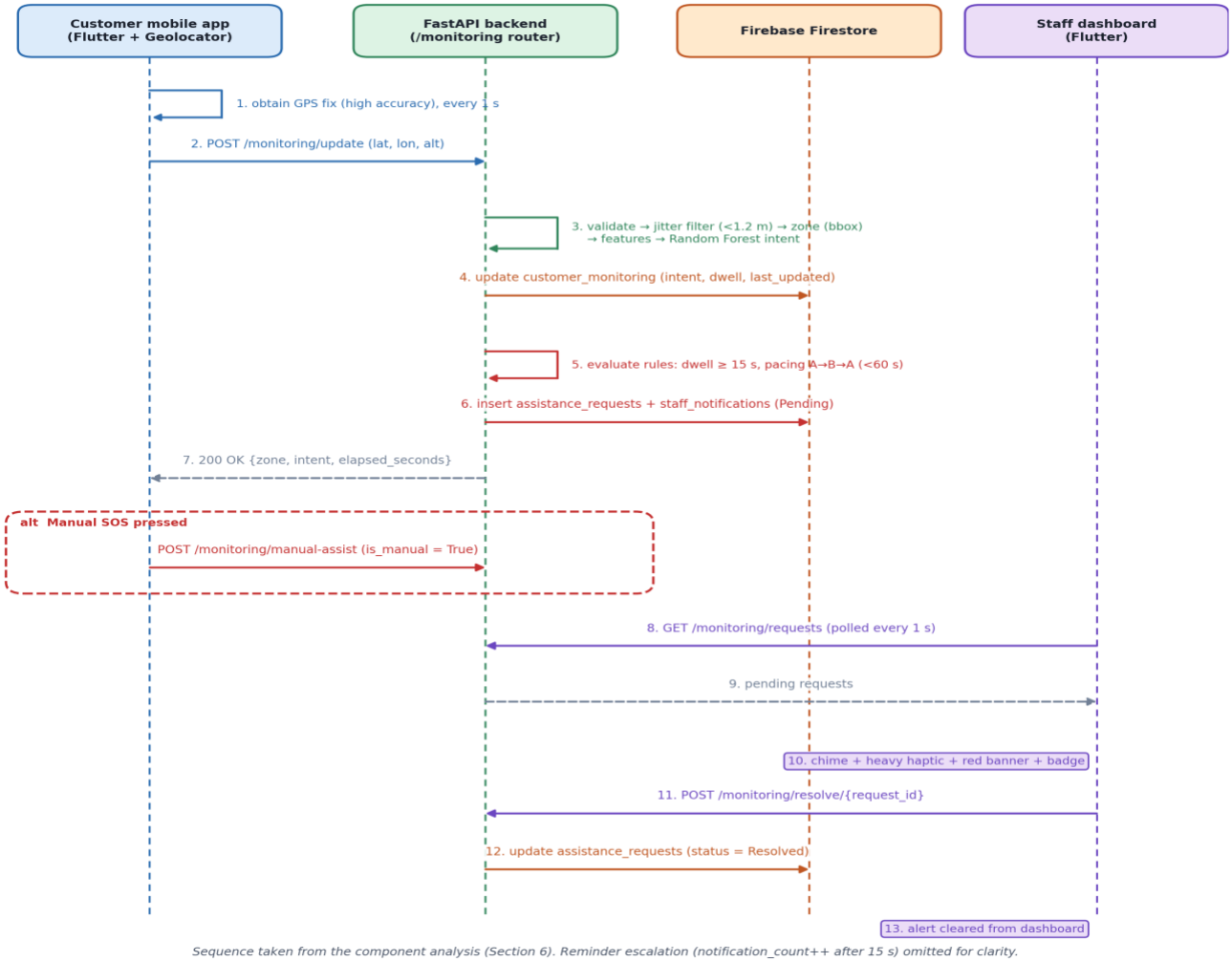
The zone engine associates incoming coordinates with configured polygons. The jitter filter is intended to reduce false transitions caused by small coordinate fluctuations. Report the exact formula and threshold, and test stationary traces, boundary traces, abrupt GPS jumps, and missing updates.

5.2.4 Random Forest and rule engine

Describe how training features are calculated and how the trained model is loaded in the backend. Include the model version and parameters. After classification, the rule engine evaluates dwell and movement patterns. The rule result should include a human-readable reason, for example 'dwell threshold reached in zone X', rather than implying certainty about a shopper's mental state.

5.2.5 Dashboard and alert resolution

The dashboard should present alert status, relevant zone, trigger reason, timestamp, and an acknowledgement/resolution control if these features exist. The supplied summary reports one-second polling, audio and haptic alerts, and Firestore resolution tracking; verify these details and include only what the deployed version demonstrates.



Sequence taken from the component analysis (Section 6). Reminder escalation (notification_count++ after 15 s) omitted for clarity.

Figure 5.1: Mobile-to-backend event sequence

5.3 Integration with Other Components

The wider system includes inventory optimisation, virtual try-on/size recommendation, and personalized marketing intelligence. Do not claim direct GPS-to-marketing or GPS-to-inventory data exchange unless implemented and demonstrated. The integration section should list shared authentication, store/product identifiers, APIs, database collections, and events that actually cross component boundaries.

Table 5.2: Integration matrix

Integration point	Expected/possible exchange	Current evidence
Mobile app → FastAPI	Location update and session context	Route identified; add request/response evidence
FastAPI → Firestore	Alert or resolution state	Reported in notes; verify collection and schema
Backend → staff dashboard	Current zone/alert/status	Add API/log/screenshot
GPS component ↔ marketing component	Aggregated zone/engagement signals only if implemented	[CONFIRM; do not assume]
GPS component ↔ inventory component	Store/zone context only if implemented	[CONFIRM; do not assume]
GPS component ↔ virtual try-on component	Customer-facing handoff only if implemented	[CONFIRM; do not assume]

5.4 Verification and Software Testing

Table 5.3: Test plan

Test ID	Scenario	Expected result	Actual result/status
T01	Valid coordinate inside a zone	Correct zone assigned	[RUN/RECORD]
T02	Coordinate outside all zones	No-zone state handled safely	[RUN/RECORD]
T03	Coordinate exactly on polygon boundary	Documented deterministic outcome	[RUN/RECORD]
T04	Small stationary GPS jitter	No repeated false transitions beyond configured filter	[RUN/RECORD]
T05	Large sudden coordinate jump	Invalid/uncertain event handled safely	[RUN/RECORD]
T06	Dwell threshold reached	Configured candidate alert generated once	[RUN/RECORD]
T07	Dwell below threshold	No dwell-triggered alert	[RUN/RECORD]
T08	A→B→A sequence within 60 seconds	Pattern detected if rule conditions hold	[RUN/RECORD]
T09	Repeated A→B→A outside time window	Pattern not triggered	[RUN/RECORD]
T10	Location permission denied	Clear UI state; no unauthorised tracking	[RUN/RECORD]
T11	Network unavailable	Graceful failure/retry; no misleading success	[RUN/RECORD]
T12	Staff acknowledges and resolves alert	Status transition persists correctly	[RUN/RECORD]
T13	Unauthenticated API request	Rejected according to policy	[RUN/RECORD]
T14	Duplicate/stale location event	Handled without duplicate alert	[RUN/RECORD]

5.5 Deployment and Operation

Record the actual deployment platform, backend host, environment variables by name only, database configuration, release process, logging/monitoring, and rollback strategy. If the component was tested locally or in a prototype, state that accurately rather than describing it as a production deployment.

5.6 Chapter Summary

Implementation claims should be supported by source files, API examples, screenshots, tests, and deployment records. The next chapter separates verified measurements from design expectations.

6. RESULTS AND VALIDATION

6.1 Research Results

The supplied research-paper excerpt reports the following classification results for the GPS-based tracking component: accuracy 91.2%, weighted precision 0.908, weighted recall 0.912, weighted F1-score 0.910, and ROC-AUC 0.895. These are reported source values, not independently reproduced here. The evaluation dataset size, class support, split protocol, and confidence intervals are not established by the available excerpt and must be added from the original experiment artefacts.

Table 6.1: Classification results reported in the supplied research paper

Metric	Reported value	Interpretation	Verification status
Accuracy	91.2%	Share of test predictions that were correct	Verify test size and output
Weighted precision	0.908	Support-weighted precision across classes	Verify class-wise report
Weighted recall	0.912	Support-weighted recall across classes	Verify class-wise report
Weighted F1-score	0.910	Support-weighted F1	Verify class-wise report
ROC-AUC	0.895	Discrimination metric as reported in paper	Conflict with another project note; verify

A separate project summary contains ROC-AUC 0.964 and PR-AUC 0.961, alongside 228/250 (91.2%) test cases and an 85 ms latency figure. Because the supplied paper excerpt instead reports ROC-AUC 0.895 and does not establish PR-AUC or the 85 ms figure in the retrieved excerpt, this draft does not combine them as though they came from the same experiment. Inspect the final notebook/logs and record which evaluation run each number belongs to.

Figure 6.1: Confusion matrix for Browsing versus Transiting

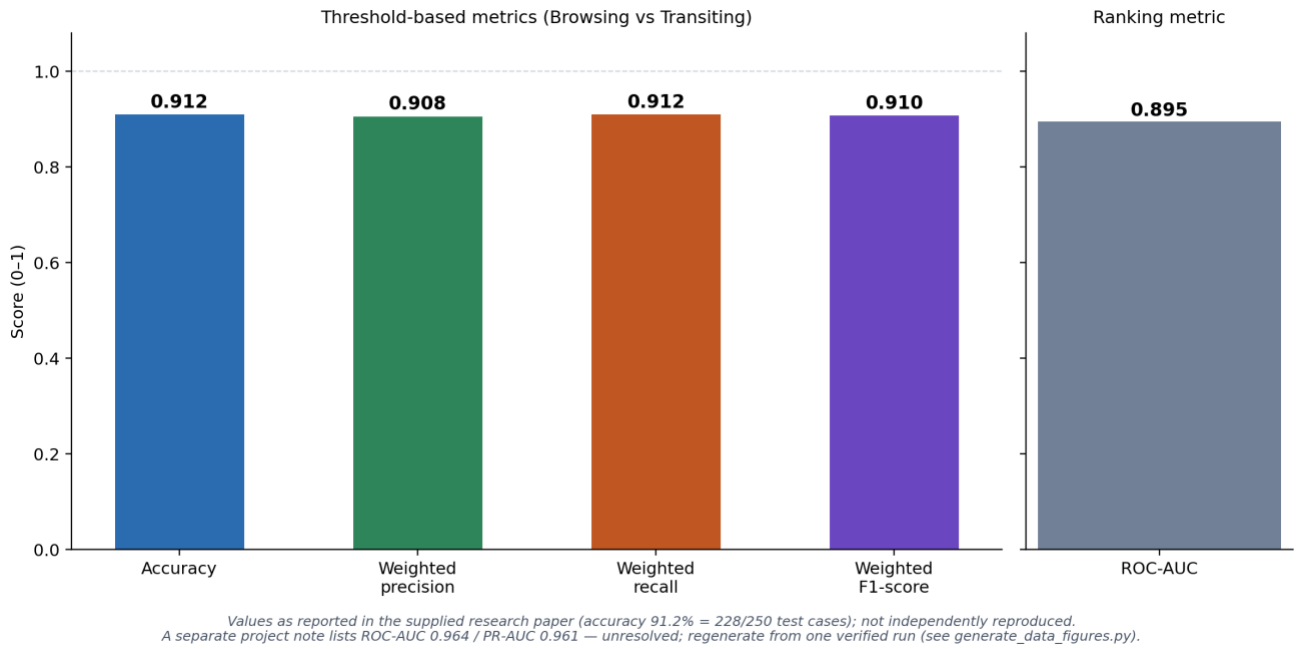


Figure 6.2: Classification metric comparison

6.2 Product and Technical Results

The component description states that the system can process zone movement, detect behaviour patterns, issue staff alerts, and provide dashboard analytics. These are intended capabilities; implementation status must be supported by screenshots or test records. Operational results should include the number of events processed, valid/invalid location updates, zone-assignment success, false alerts, dropped requests, and end-to-end latency.

Table 6.2: Operational results template

Measure	Value	Method/source	Status
Total location updates evaluated	[ADD]	Server logs/test dataset	Missing
Valid zone assignment rate	[ADD]	Zone test output	Missing
Assistance alerts generated	[ADD]	Event log	Missing
False/duplicate alerts	[ADD]	Annotated scenarios	Missing
Median alert latency	[ADD] ms	Timestamp difference	Missing
95th-percentile latency	[ADD] ms	Timestamp difference	Missing
API error rate	[ADD] %	Request/error logs	Missing
Alert resolution success	[ADD] %	Workflow test results	Missing

6.3 User or Stakeholder Validation

The component explanation records qualitative feedback that real-time tracking and alerts are useful, staff may respond faster, and managers appreciated behaviour analytics. It also records suggestions to improve indoor GPS accuracy and make alert priorities clearer. The available material does not specify participant count, recruitment, questionnaire wording, scoring, or analysis. Present these comments as informal feedback only unless the underlying study documentation supports a formal user evaluation.

Table 6.3: Feedback summary from component explanation

Reported feedback	Implication	Evidence limitation
Tracking and alerts are useful	Potential value in timely support	Participant count and collection method missing
Helps staff respond faster	Workflow may improve responsiveness	No measured response-time comparison provided
Managers appreciated analytics dashboard	Dashboard may support awareness	No usability score or task metrics provided
Improve GPS accuracy indoors	Location quality is a key limitation	Needs device/indoor validation
Clearer alert priority indicators	Prioritisation may improve usability	Needs usability test and revised UI evidence

6.4 Integration Results

Integration success should be reported using actual end-to-end test evidence: mobile update accepted by backend, zone and behaviour state computed, alert persisted, dashboard updated, and staff action reflected in the stored status. The one-second dashboard polling interval and Firestore resolution status are reported in the component summary but must be confirmed with logs or code.

6.5 Summary of Findings

The supplied paper provides promising reported classification metrics, but a complete validation requires dataset documentation, confusion-matrix counts, operational latency, alert quality, and reproducible end-to-end tests. Indoor location accuracy and interpretation of movement patterns remain central limitations.

6.6 Chapter Summary

This chapter separates model metrics from product operation and user feedback. It deliberately marks missing results rather than fabricating evidence.

7. DISCUSSION

7.1 Interpretation of Findings

If reproduced, a 91.2% accuracy and weighted F1 of 0.910 would suggest that the classifier separates the evaluated Browsing and Transiting labels reasonably well on that test set. However, accuracy alone cannot establish real-world assistance quality. The class distribution, annotation reliability, leakage risk, and representativeness of indoor trajectories must be examined. Even a correct movement classification may not mean that the customer needs help.

7.2 Comparison with Literature, Baselines and Existing Solutions

The component should be compared with simple baselines using the same test split, and related work should be compared by task and sensing modality rather than by headline percentages alone. GPS-only tracking, BLE/Wi-Fi positioning, camera analytics, and RTLS systems have different hardware, privacy, and accuracy trade-offs. Avoid direct numerical comparison when datasets, labels, and evaluation protocols differ.

Table 7.1: Evidence-based comparison framework

Comparison dimension	This component	Alternative systems	Caution
Hardware	Ordinary mobile GPS is intended	BLE/Wi-Fi/RTLS/cameras may require infrastructure	Cost and accuracy differ
Behaviour label	Browsing vs Transiting	Other studies may predict paths, visits or dwell	Labels are not interchangeable
Assistance workflow	Rule-triggered staff alert	Some systems focus only on analytics or navigation	Measure full workflow separately
Privacy profile	Location data collected with mobile permission	Camera/beacon systems collect different signals	All require governance
Validation	Paper reports classifier metrics; operational evidence incomplete here	Published studies use varied datasets	Do not compare raw accuracy without protocol alignment

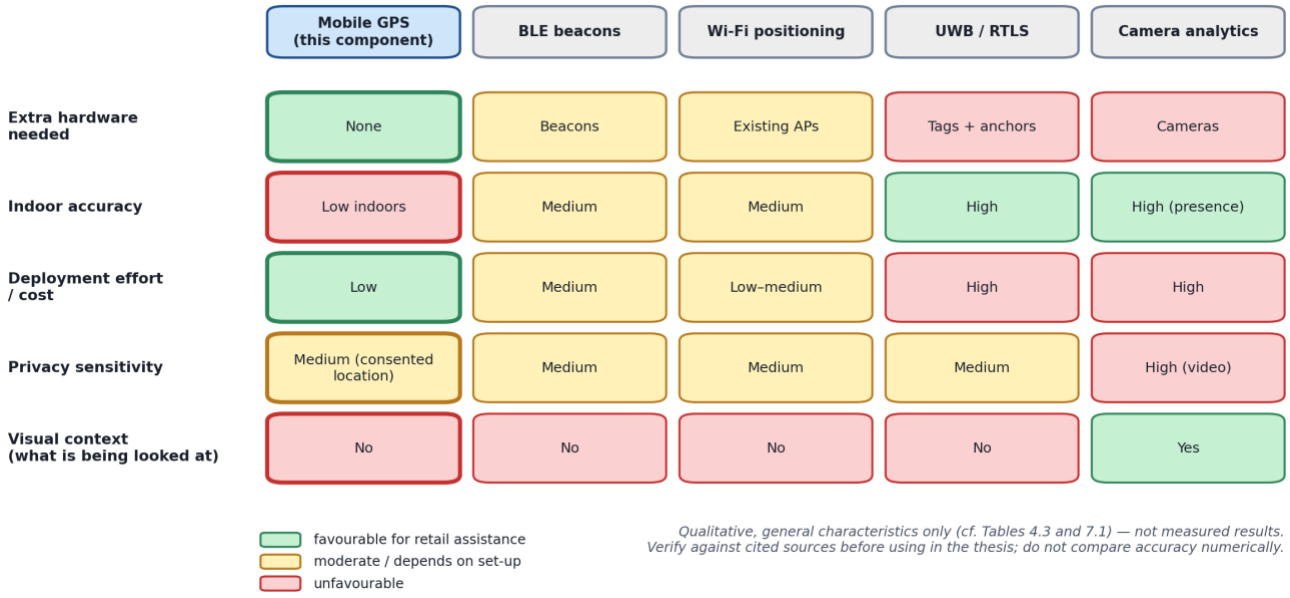


Figure 7.1: Comparison of positioning/analytics approaches

7.3 Research Contribution

The intended research contribution is a practical integration of mobile location acquisition, zone assignment, noise filtering, movement classification, and rule-based assistance notification in an apparel retail context. The contribution should not be overstated as a new geofencing algorithm or a definitive method for detecting emotion. Its strength depends on reproducible evaluation and evidence that the end-to-end system functions as described.

7.4 Product Contribution

The product contribution is a potential staff-support tool that converts selected location and movement events into actionable prompts. A useful interface should provide a transparent reason for each alert, allow staff to dismiss or resolve it, and avoid overwhelming employees with repetitive notifications. Aggregate zone analytics may also help managers understand traffic patterns, subject to consent and data minimisation.

7.5 Limitations and Threats to Validity

Table 7.2: Limitations and validity threats

Threat	Possible effect	Mitigation / future work
GPS drift indoors	Wrong zone and false transitions	Use uncertainty thresholds and evaluate alternative positioning
Dataset not representative	Poor generalisation to real customers/devices	Collect consented, diverse real-world traces
Label ambiguity	Model learns inconsistent behaviour labels	Define annotation protocol and measure agreement
Data leakage	Inflated test metrics	Split by session/person/device and freeze test set
Heuristic thresholds	False alerts or missed assistance opportunities	Tune on validation set; conduct sensitivity analysis
Movement ≠ intent	Intrusive or incorrect interpretation	Use neutral alert language and human judgement
Battery/background limits	Missing updates or reduced adoption	Measure resource use and test lifecycle behaviour
Metric inconsistency	Incorrect thesis conclusions	Reconcile ROC-AUC and latency from final experiment artefacts

7.6 Implications

For retailers, the system may provide a low-cost supplementary signal for staff allocation and assistance. For customers, benefits depend on transparent consent and respectful use. For researchers, the work illustrates the need to evaluate not only model classification but also location uncertainty, alert usefulness, and human response. For developers, modular APIs and explicit state transitions support maintenance and integration.

7.7 Chapter Summary

The findings should be interpreted as evidence about the tested system and dataset only. Deployment claims require additional field evaluation and privacy review.

8. COMMERCIALISATION AND SUSTAINABILITY

8.1 Value Proposition and Target Users

The potential value proposition is to help apparel retailers notice candidate assistance situations earlier without installing a camera system or dedicated beacon network. Primary users are store associates and managers; customers are affected stakeholders whose location data must be collected transparently and minimally.

Table 8.1: Stakeholders and value proposition

Stakeholder	Potential value	Condition for value
Store associates	Timely prompts to offer help	Alerts must be accurate and not excessive
Store managers	Zone activity and response overview	Dashboard must present meaningful, privacy-safe summaries
Customers	Potentially faster support and less friction	Clear consent, control and respectful messaging
Retail operations	Evidence for staffing and service improvement	Reliable field data and integration with store processes
IT/security team	Manageable mobile/backend architecture	Access control, retention and monitoring must be documented

8.2 Market and Competitive Position

The proposed positioning is a lightweight assistance layer for retailers who want location-aware support but may not be ready to deploy dedicated indoor-positioning infrastructure. It should be compared against manual assistance processes, camera analytics, BLE/Wi-Fi positioning, and retail analytics platforms. No market-size or competitor-price claim is included because no market study was provided; add verified sources if this section is required for assessment.

8.3 Business or Sustainability Model

Potential delivery models include a per-store subscription, a software licence with support, or inclusion in a wider smart-retail platform. These are proposals rather than a confirmed business model. Pricing should account for onboarding, zone configuration, backend hosting, support, privacy compliance, and device/network costs. Pilot evaluation should demonstrate operational value before a commercial return-on-investment claim is made.

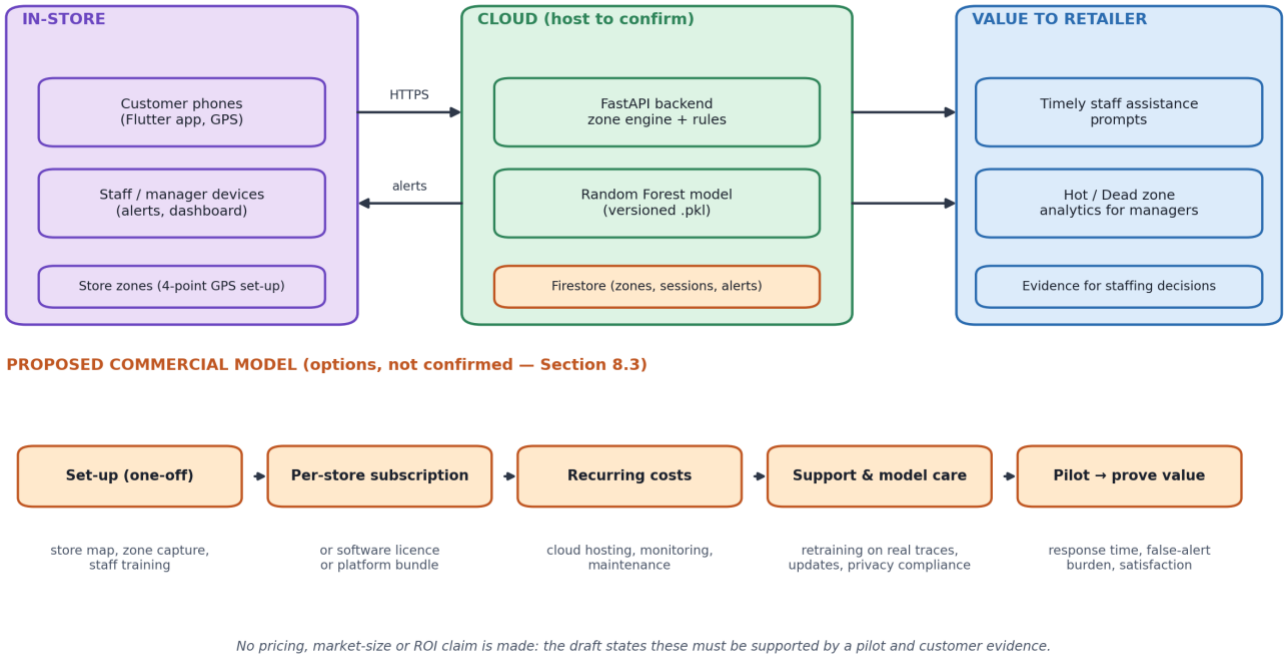


Figure 8.1: Deployment and Commercial Operating Model

Table 8.2: Indicative cost and revenue categories

Category	Examples	Evidence needed
Setup	Store map, zone configuration, staff training	Time/cost estimate
Recurring costs	Cloud hosting, monitoring, maintenance	Provider pricing and expected usage
Support	Troubleshooting, model monitoring, updates	Support plan and staffing assumptions
Revenue option	Per-store subscription or platform bundle	Customer interviews and willingness-to-pay evidence
Value measures	Response time, false-alert burden, customer satisfaction	Pilot baseline and post-deployment measurements

8.4 Scalability, Intellectual Property and Compliance

Scaling requires configurable zones per store, rate limits, robust session management, monitoring, model-version control, and a strategy for poor connectivity. Intellectual-property ownership must follow the institution and group project rules; identify third-party libraries and licences. Because location traces can be sensitive, assess applicable Sri Lankan data-protection law, institutional requirements, consent, purpose limitation, retention, access rights, and security safeguards before commercial deployment.

8.5 Contribution to Sustainable Development Goals

The project may relate to SDG 9 (Industry, Innovation and Infrastructure) through applied digital infrastructure and SDG 12 (Responsible Consumption and Production) only if evidence supports reduced waste or improved operational efficiency. It may also support SDG 8 (Decent Work and Economic Growth) if the system improves service operations without unfairly monitoring staff. These are potential alignments, not measured impacts; avoid claiming SDG achievement without evidence.

Table 8.3: Sustainability and SDG mapping

Area	Potential alignment	Evidence/qualification
SDG 9	Digital innovation for retail operations	Demonstrate reliable deployment and useful outcomes
SDG 8	Support staff service workflows	Assess workload, fairness and staff experience
SDG 12	Potentially more efficient retail operations	No waste reduction demonstrated; measure before claiming
Environmental impact	Avoid dedicated camera/beacon hardware in initial design	Compare actual device, network and cloud resource use
Social/privacy impact	Transparent, limited-purpose location tracking	Consent, retention and access safeguards required

8.6 Chapter Summary

Commercial potential depends on demonstrating that assistance alerts are useful, accurate enough, privacy-respecting, and operationally affordable. The current draft identifies options rather than asserting proven market demand.

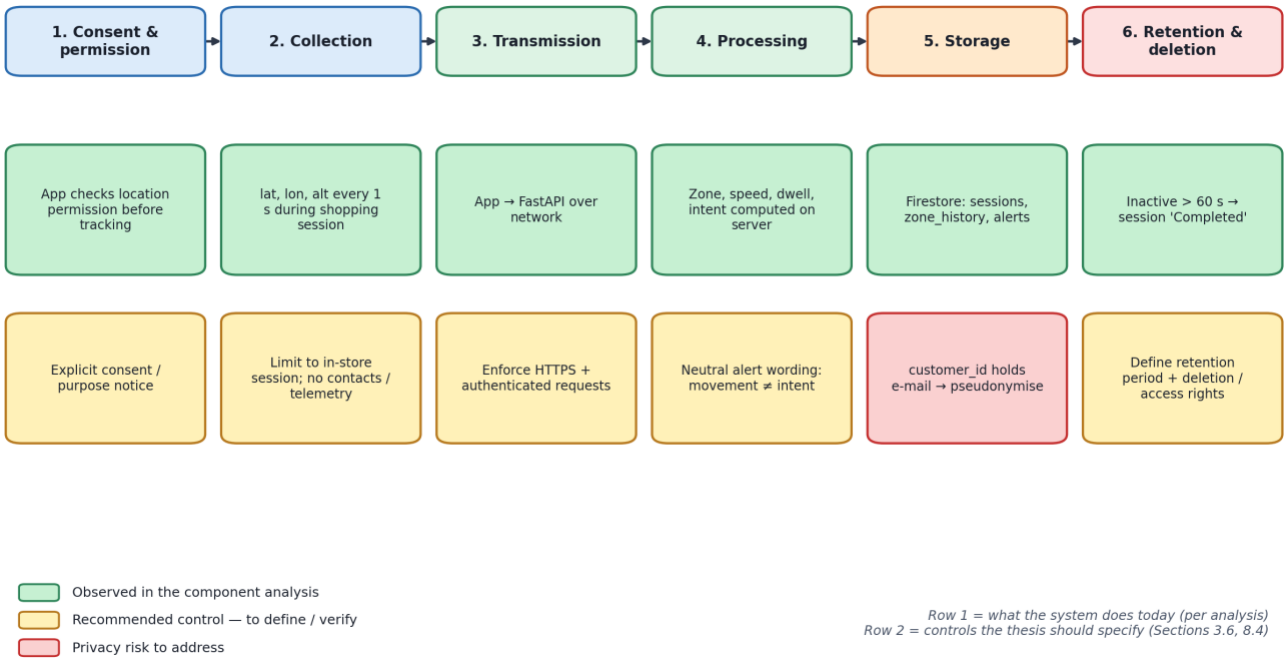


Figure 8.2: Privacy- aware Data Life cycle

9. CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This thesis draft presents a GPS-based customer behaviour tracking and real-time staff assistance component for an AI-Based Smart Retail Recommendation System for Apparel. The proposed workflow combines mobile location updates, zone assignment, jitter filtering, Random Forest classification, configurable dwell/movement rules, and a staff-facing alert process. The supplied research paper reports promising classifier metrics, but full validation requires confirmed dataset details, reproducible evaluation outputs, and operational tests. Indoor GPS limitations, the ambiguity of behavioural labels, battery/background restrictions, and privacy obligations constrain the claims that can be made.

9.2 Achievement of Objectives

Table 9.1: Objective achievement matrix

Objective	Evidence currently available	Draft status
Define zones and map location updates	Point-in-polygon and jitter filtering described	Partially documented; code/tests required
Classify browsing vs transiting	Paper reports classification metrics	Reported; reproduce and document dataset/split
Detect assistance opportunities	Dwell and A→B→A rules described	Rules require unit-test evidence
Notify staff and track resolution	Dashboard/audio-haptic/Firestore behaviour described in notes	Verify implementation and end-to-end logs
Evaluate limitations and responsible use	Indoor GPS, battery and privacy limitations identified	Field and privacy validation incomplete

9.3 Individual Research and Product Contributions

The contribution should be finalised from repository history and supervisor guidance. A defensible claim is that the component explores an integrated mobile-GPS-to-staff-alert workflow for retail support. The final thesis should state precisely which mobile, backend, ML, dashboard, testing, and documentation tasks were personally completed, and which services were shared or contributed by other group members.

9.4 Future Directions

Future work should prioritise: (1) testing in real store layouts with informed consent; (2) comparing GPS with BLE, Wi-Fi, or hybrid positioning; (3) collecting a diverse, labelled dataset and evaluating cross-device generalisation; (4) calibrating and validating dwell/movement thresholds; (5) measuring alert usefulness, response time, and false-alert burden; (6) optimising background tracking and battery consumption; (7) improving alert priority and explanation; (8) strengthening retention, deletion, and privacy controls; and (9) conducting a controlled pilot with staff and customer feedback.

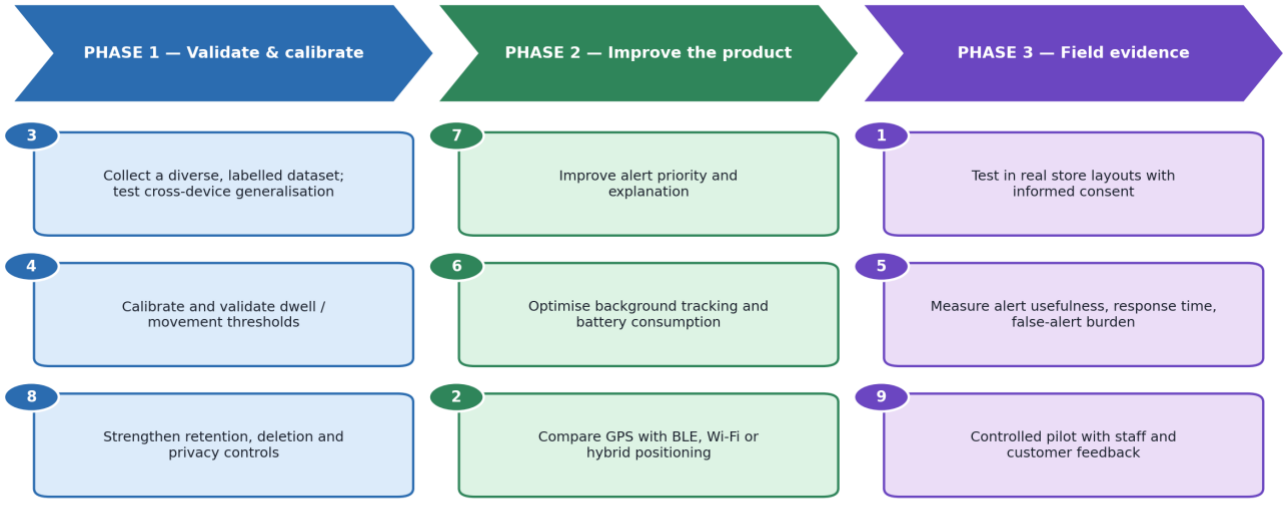


Figure 9.1: Future work roadmap

9.5 Final Summary

The component offers a plausible approach to supplementing retail staff awareness with mobile location signals. Its strongest contribution will be established by reproducible model evaluation and evidence that the end-to-end assistance workflow works reliably and respectfully in a real retail environment.

LIST OF PLANNED FIGURES

Table F.1: Figure register

Figure	Title
1.1	Overall smart retail system context
1.2	Individual component boundary and ownership
2.1	Conceptual framework and research gap
3.1	Research methodology workflow
4.1	Component architecture
4.2	Zone transition and assistance decision flow
5.1	Mobile-to-backend event sequence
6.1	Confusion matrix for Browsing versus Transiting
6.2	Classification metric comparison
6.3	Alert latency distribution (after logs are collected)
6.4	Zone-assignment and rule-trigger test results (after tests are run)
7.1	Comparison of positioning/analytics approaches
8.1	Proposed deployment and commercial operating model
8.2	Privacy-aware data lifecycle
9.1	Future-work roadmap

REFERENCES

The following sources are included as initial references based on the supplied paper and relevant contextual research. Check exact authors, titles, volume/issue/pages, DOI, access date, and institutional citation style against the original publications before submission.

[1] L. M. H. Alahakoon et al., “An AI Driven Smart Retail Recommendation System,” supplied project research paper, 2026. [Confirm full author list, venue, pagination, and DOI from the original PDF.]

[2] “Geofencing in location-based behavioral research: Methodology, challenges, and implementation,” Behavior Research Methods, 2023. doi: 10.3758/s13428-023-02213-2.

[3] “Robust and affordable retail customer profiling by vision and radio beacon sensor fusion,” Pattern Recognition Letters, vol. 81, pp. 30–40, Oct. 2016. doi: 10.1016/j.patrec.2016.02.010. [Verify author list from publisher record.]

[4] “A business application of RTLS technology in Intelligent Retail Environment: Defining the shopper's preferred path and its segmentation,” Journal of Retailing and Consumer Services. [Verify full bibliographic metadata from publisher.]

[5] N. E. M. Mohamed, M. A. Shafea, and M. M. Abdelhakam, “Anomaly-resilient geofencing and predictive navigation in IoT environments using machine learning and federated learning for metaverse workplaces and smart shopping malls,” Scientific Reports, vol. 16, art. 2639, 2026. doi: 10.1038/s41598-025-33856-0.

[6] Parliament of the Democratic Socialist Republic of Sri Lanka, Personal Data Protection Act, No. 9 of 2022, Colombo, Sri Lanka, 2022. [Verify current commencement and applicable provisions using official sources.]

[7] Scikit-learn developers, “RandomForestClassifier documentation.” [Use the official documentation URL and access date for the version used.]

[8] Flutter developers, “Flutter documentation.” [Use official documentation and version/access date.]

[9] FastAPI developers, “FastAPI documentation.” [Use official documentation and version/access date.]

[10] Google, “Geolocator plugin / Android location permissions documentation.” [Replace with exact official documentation used during implementation.]

GLOSSARY

Term	Definition
Geofence	Virtual geographic boundary used to associate location observations with a defined area.
Point-in-polygon	Geometric test for whether a point lies inside a polygon.
Dwell time	Elapsed time that a session is estimated to remain in a zone.
Browsing	One of the classifier's target labels; exact annotation definition must be documented.
Transiting	One of the classifier's target labels; exact annotation definition must be documented.
Jitter	Small fluctuations in location observations that can cause false zone changes.
False positive	An alert or class prediction that indicates an event when the reference label says otherwise.
Alert latency	Elapsed time from a qualifying trigger to the alert becoming visible to staff.
Firestore	Cloud-hosted document database in the Firebase platform.
Human-in-the-loop	A design in which a human reviews or acts on system output rather than treating it as an unquestionable decision.

APPENDICES

Appendix A — Dataset Card Template

Table A.1: Dataset card

Field	Complete before submission
Dataset name/version	[ADD]
Origin and collection date	[ADD]
Collection setting	[ADD]
Number of users/sessions/updates	[ADD]
Coordinate and timestamp units	[ADD]
Features and labels	[ADD]
Missing/outlier handling	[ADD]
Train/validation/test split	[ADD]
Consent and ethics approval	[ADD]
Known biases and limitations	[ADD]

Appendix B — API Documentation Template

Document the actual /monitoring/update route: HTTP method, endpoint URL without secrets, authentication requirement, request JSON example with fictional coordinates, response schema, status codes, validation rules, rate limits, and error behaviour. Never include live customer coordinates or access tokens.

Appendix C — Test Evidence

Insert actual test screenshots, console output, test reports, API responses, and staff-dashboard evidence. Each item should include a test ID, date, build/version, expected result, actual result, and pass/fail status. Use fictional or anonymised data.

Appendix D — User Feedback Instrument

If formal user evaluation was conducted, include the approved questionnaire, participant information and consent materials, anonymised participant profile, response counts, scoring approach, and analysis. If only informal comments were collected, label them as informal feedback and do not report statistical conclusions.

Appendix E — Model Configuration and Results

Include the exact feature list, label mapping, Random Forest parameters, random seed, library versions, train/validation/test split, class support, confusion matrix, classification report, ROC-AUC method, and the final saved model version.

Appendix F — Figure Creation Checklist

Table F.2: Figure completion checklist

Figure group	Required action
Architecture and context figures	Create in draw.io/Visio from verified architecture and ownership
Methodology and sequence diagrams	Match actual implementation and error paths
Model charts	Generate from final saved evaluation outputs; include counts and labels
Latency and rule charts	Generate only after running and recording tests
Dashboard and mobile screenshots	Capture from real application with personal/customer data removed
Privacy lifecycle	Map actual collection, processing, storage, retention, and deletion